Simulation of a two-link planar anthropomorphic simulator

Javier Castro Santos

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Abstract

This paper presents the modeling, simulation, and control of a two-link planar anthropomorphic manipulator with a MATLAB/Simulink interface. The main purpose is to study the dynamics of the nonlinear robot and to realize a computed torque control (CTC) scheme that can successfully follow the desired joint trajectories. The mathematical formulation is derived based on the non-dimensional dynamic model introduced in the project summary with a structure of inertia matrix, Coriolis and centrifugal terms, auxiliary control inputs.

A complete dynamic simulation in Simulink is formulated that includes the inverse dynamics of both the manipulator, CTC subsystem, and the generation of step-based trajectory with a rotation limit of 30°. Some simulations were performed for different values of the non-dimensional tip-mass parameter $\mu = 0,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 2,\ 10$ to assess controller sensitivity and robustness.

They reveal that for most parameter values, the computed-torque controller effectively linearizes and decouples dynamics, while high values of μ cause torque requirements to increase dramatically. The numerical difficulties for $\mu = 0$ are also addressed. The work emphasizes the significance of parameter matching between model and controller as well as the constraint of the system when the inertia matrix tends to singularity.

1 Introduction

The nonlinear coupled dynamics of industrial robotic manipulators demands precise modeling and robust control techniques. The aim of this work is the simulation and control of a two-link planar anthropomorphic manipulator, which is a common benchmark model used for purposes of nonlinear robot dynamics. The set of the non-dimensional dynamic equations based on the Euler–Lagrange formulation is given in the project brief. These equations express the joint torques T_1, T_2 as functions of the joint accelerations $\ddot{\theta}_1, \ddot{\theta}_2$:

$$M^*(\theta_2)\begin{bmatrix}\ddot{\theta}_1\\ \ddot{\theta}_2\end{bmatrix} + C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2) = \begin{bmatrix}T_1\\ T_2\end{bmatrix}. \tag{1}$$

where $M^*(\theta_2)$ is the non-dimensional inertia matrix and $C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2)$ contains the Coriolis and centrifugal effects. A key parameter in the model is the non-dimensional tip mass,

$$\mu = \frac{m_{\text{tip}}}{m_{\text{link}}} \tag{2}$$

which strongly influences the dynamic behaviour.

A central goal of the assignment is the implementation of a computed–torque controller (CTC), which cancels the nonlinear dynamics and forces the system to behave like two uncoupled linear second-order systems:

$$v_1 = \ddot{\theta}_{1d} + 2\zeta_1\omega_1(\dot{\theta}_{1d} - \dot{\theta}_1) + \omega_1^2(\theta_{1d} - \theta_1),$$

$$v_2 = \ddot{\theta}_{2d} + 2\zeta_2\omega_2(\dot{\theta}_{2d} - \dot{\theta}_2) + \omega_2^2(\theta_{2d} - \theta_2). \tag{4}$$

The required torques are computed using

$$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = M^*(\theta_2) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2). \tag{5}$$

The Simulink model developed includes trajectory generation, auxiliary control computation, non-dimensional torque calculation, and inverse dynamics integration. A particular requirement of the brief is that the servo motors cannot rotate more than 30° at a time; therefore, achieving a rotation of 240° requires eight sequential 30° step inputs. The performance of the controller is evaluated for multiple values of μ.

The aims of this project are:

- To construct a fully functional Simulink model of the two-link manipulator.
- To implement the computed–torque control law as specified.
- To analyse auxiliary control signals v_1, v_2 and torque signals T_1, T_2 under step-based demands.
- To examine the sensitivity of the system to variations in the parameter μ.

2 System Modelling

The two-link planar anthropomorphic manipulator under study has identical links with unit length and unit mass. At the tip of link 2 there is a point mass 2μ, where the non-dimensional parameter

$$\mu = \frac{m_{\text{tip}}}{m_{\text{link}}}, \tag{6}$$

is used throughout the formulation. The positions, velocities and accelerations of the joints are

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \qquad \dot{\boldsymbol{\theta}} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, \ \ddot{\boldsymbol{\theta}} = \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}. \tag{7}$$

Verification of the Dynamic Equations Using the Lagrangian Method

The equations of motion provided in the project brief were verified using the Euler–Lagrange formulation. For a two–link manipulator with joint coordinates $q = [\theta_1, \theta_2]^T$, the Lagrangian is

$$L = T - V,$$

where T is the total kinetic energy and V the gravitational potential energy. Since the motion occurs in a horizontal plane, V is constant and $\partial V/\partial q_i = 0$.

Applying the Euler–Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i, \tag{8}$$

with $Q_i = T_i$, yields the nonlinear manipulator dynamics

$$M^*(\theta_2)\ddot{\boldsymbol{\theta}} + C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2) = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}, \tag{9}$$

which match the non–dimensional forms (7)–(11) used in this report. This confirms that the dynamic model implemented in Simulink is consistent with the Lagrangian derivation.

2.1 Non-Dimensional Dynamic Equations

Employing the Euler–Lagrange equations, in standard robotic terms, the project description provides non-dimensionalized equations of motion:

$$M^*(\theta_2)\,\ddot{\boldsymbol{\theta}} C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2) = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}, \tag{10}$$

where $M(\theta_2)$ is an inertia matrix and $C(\theta_2, \dot{\theta}_1, \dot{\theta}_2)$ is the coriolis and centripetal matrix. These terms describe the nonlinear coupling of the two joints.

2.2 Inertia Matrix

The non-dimensional inertia matrix provided in the note was given by

$$M^*(\theta_2) = \begin{bmatrix} 1 + \mu + 2\mu\cos\theta_2 & \mu\cos\theta_2 + \frac{\mu}{2} \\ \mu\cos\theta_2 + \frac{\mu}{2} & \frac{\mu}{2} \end{bmatrix}. \tag{11}$$

What we see is that stronger flanking links have two leading sites of greatest weight in I IV. Of course, one can also argue with μ as before.

For $\mu = 0$ the inertia matrix is degenerate by numerical simulation, this will require a special treatment (see Section 6).

2.3 Coriolis and Centrifugal Terms

The velocity dependent non-linear terms are organized into

$$C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2) = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}, \tag{12}$$

with components

$$C_1 = -\mu(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2)sin(\theta_2), \qquad C_2 = \mu\left(\dot{\theta}_1^2\right)\sin\theta_2. \tag{13}$$

These coupling effects induced by these nonlinear terms are to be canceled out by the control law so that the trajectory is tracked over time.

2.4 Summary

The full non-dimensional dynamic model employed in simulationand control is given by

$$M^*(\theta_2)\,\ddot{\boldsymbol{\theta}} C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2) = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix}. \tag{14}$$

This formulation forms the basis for the computed–torque control approach discussed in Section 4.

Identification of Dynamic Terms

From the non–dimensional model, the physical contributions are:

- **Inertia matrix** $M^*(\theta_2)$**:** Determines how joint accelerations couple and how the robot resists motion.
- **Coriolis and centrifugal terms** $C^*(\theta_2, \dot{\theta})$**:** Velocity–dependent forces, such as $-\mu(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2)\sin\theta_2$, which couple the joints.
- **Gravity term:** Absent in this system because the arm moves in a horizontal plane ($g = 0$), so the potential energy is constant.

These terms are essential because computed–torque control relies on their exact cancellation to achieve linear, decoupled closed–loop behaviour.

3 Control Design

Justification for Setting $g = 0$

The manipulator operates in a horizontal plane, so gravity produces no torque about either joint. The potential energy V is constant and $\partial V/\partial\theta_i = 0$, which means the gravity vector does not contribute to the dynamic equations. Therefore, setting $g = 0$ is physically valid and simplifies the non–dimensional model without affecting the robot behaviour.

The objective of the control strategy is to ensure that the two–link manipulator follows a given joint trajectory as accurately as possible despite its nonlinear, coupled dynamics. The control method used in this work is the computed–torque controller (CTC), which cancels the nonlinear dynamics and forces the closed–loop system to behave as two independent linear second–order systems.

3.1 Control Objective

Let the desired joint trajectories be

$$\boldsymbol{\theta}_d = \begin{bmatrix} \theta_{1d} \\ \theta_{2d} \end{bmatrix}, \qquad \dot{\boldsymbol{\theta}}_d,\ \ddot{\boldsymbol{\theta}}_d. \tag{15}$$

Define the tracking errors

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \boldsymbol{\theta}_d - \boldsymbol{\theta}, \qquad \dot{\mathbf{e}} = \dot{\boldsymbol{\theta}}_d - \dot{\boldsymbol{\theta}}. \tag{16}$$

The objective is to find torque inputs $\mathbf{T}$ such that $e_i(t) \to 0$ as $t \to \infty$, for $i = 1, 2$.

3.2 Auxiliary Control Inputs

In computed–torque control, the nonlinear robot dynamics are cancelled by introducing the auxiliary control inputs (v_1, v_2):

$$v_i = \ddot{\theta}_{id} + 2\zeta_i\omega_i(\dot{\theta}_{id} - \dot{\theta}_i) + \omega_i^2(\theta_{id} - \theta_i), \qquad i = 1, 2, \tag{17}$$

where (ζ_i, ω_i) are the desired closed–loop damping ratios and natural frequencies.

3.3 Computed–Torque Law

The control torques are obtained by

$$\mathbf{T} = M^*(\theta_2) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2), \tag{18}$$

where M^* is the non–dimensional inertia matrix and C^* contains the Coriolis and centrifugal effects. In expanded form:

$$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} 1 + \mu + 2\mu\cos\theta_2 & \mu\cos\theta_2 + \frac{\mu}{2} \\ \mu\cos\theta_2 + \frac{\mu}{2} & \frac{\mu}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} -\mu(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2)\sin\theta_2 \\ \mu(\dot{\theta}_1^2)\sin\theta_2 \end{bmatrix}. \tag{19}$$

When substituted into the robot dynamics, this law yields the closed–loop second–order error dynamics:

$$\ddot{e}_i + 2\zeta_i\omega_i\dot{e}_i + \omega_i^2 e_i = 0, \qquad i = 1, 2, \tag{20}$$

which are linear, decoupled, and asymptotically stable.

3.4 Effect of External Torques (Exercise ii)

In the general case, the total joint torques may differ from the nominal computed–torque inputs. Let

$$\mathbf{T}_c = \begin{bmatrix} T_{1c} \\ T_{2c} \end{bmatrix} \qquad \text{and} \qquad \Delta\mathbf{T} = \begin{bmatrix} \Delta T_1 \\ \Delta T_2 \end{bmatrix} \tag{21}$$

denote the baseline computed torques and unexpected external torques, respectively. The total torques are therefore

$$\mathbf{T} = \mathbf{T}_c + \Delta\mathbf{T}. \tag{22}$$

Substituting this into the robot dynamics and repeating the CTC derivation leads to the forced second–order error system:

$$\ddot{\mathbf{e}} + 2\omega_n\dot{\mathbf{e}} + \omega_n^2\mathbf{e} = H(\theta_2)\,\Delta\mathbf{T}, \tag{23}$$

where $H(\theta_2)$ is a matrix whose entries depend on the manipulator inertia parameters. When $\Delta\mathbf{T} = 0$, the homogeneous error dynamics are recovered. When $\Delta\mathbf{T} \neq 0$, the error does not converge exactly to zero and tracking performance is degraded.

3.5 30° Step Constraint

A practical constraint of the system is that the servo motor cannot rotate more than 30° at once. Thus a commanded rotation of 240° must be decomposed into eight consecutive steps:

$$0^\circ \;\rightarrow\; 30^\circ \;\rightarrow\; 60^\circ \;\rightarrow\; \cdots \;\rightarrow\; 240^\circ.$$

At each step, the new demanded angle $(\theta_{id}, \dot{\theta}_{id}, \ddot{\theta}_{id})$ produces updated auxiliary inputs and torques. The simulation results corresponding to these stepped motions are presented in Section 5.

4 Simulink Implementation

A complete Simulink model was developed to simulate both the nonlinear dynamics of the manipulator and the computed–torque control law described in Sections 3 and 4. The full model consists of four main subsystems:

1. Demand subsystem (generation of the desired trajectory),
2. Auxiliary control subsystem (computation of v_1 and v_2),
3. Non-dimensional torque subsystem (evaluation of the torque law),
4. Plant subsystem implementing the non-dimensional inverse dynamics.

A block diagram of the overall structure is shown in Figure 1.

4.1 Demand Subsystem

The demand subsystem generates the desired joint positions $\theta_{1d}(t)$ and $\theta_{2d}(t)$. For exercises involving unrestricted motion, step or constant references were used. For the 30° motion constraint (Exercise (v)), the subsystem outputs a sequence of eight successive 30° increments to achieve a total rotation of 240°.

The subsystem also provides the required derivatives:

$$\dot{\theta}_{id}(t) = 0, \qquad \ddot{\theta}_{id}(t) = 0, \tag{24}$$

except at the discontinuities of the step input. A placeholder for the subsystem structure is shown in Figure 2

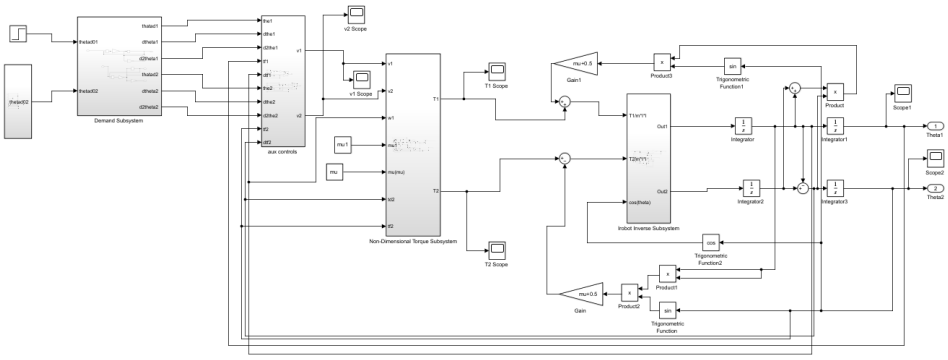


Figure 1: Top-level structure of the Simulink model.

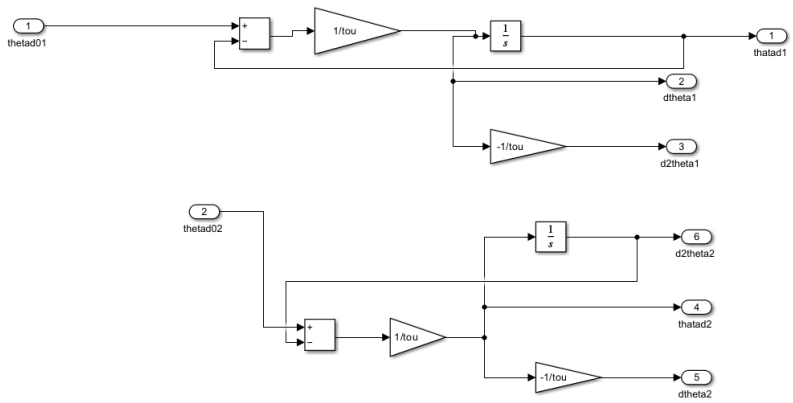


Figure 2: Demand Subsystem.

4.2 Auxiliary Control Subsystem

This subsystem implements the auxiliary control inputs defined by the computed–torque control law:

$$v_1 = \ddot{\theta}_{1d} + 2\zeta_1\omega_1(\dot{\theta}_{1d} - \dot{\theta}_1) + \omega_1^2(\theta_{1d} - \theta_1),$$
$$v_2 = \ddot{\theta}_{2d} + 2\zeta_2\omega_2(\dot{\theta}_{2d} - \dot{\theta}_2) + \omega_2^2(\theta_{2d} - \theta_2). \tag{25}$$

The subsystem receives the desired trajectory and the measured joint states, and outputs v_1 and v_2. Gain blocks implement the feedback coefficients $2\zeta_i\omega_i$ and ω_i^2. A schematic placeholder is shown in Figure 3.

4.3 Non-Dimensional Torque Subsystem

This subsystem evaluates the full computed–torque control law:

$$\mathbf{T} = M^*(\theta_2)\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2), \tag{26}$$

using the expressions for M^* and C^* defined in Section 3.

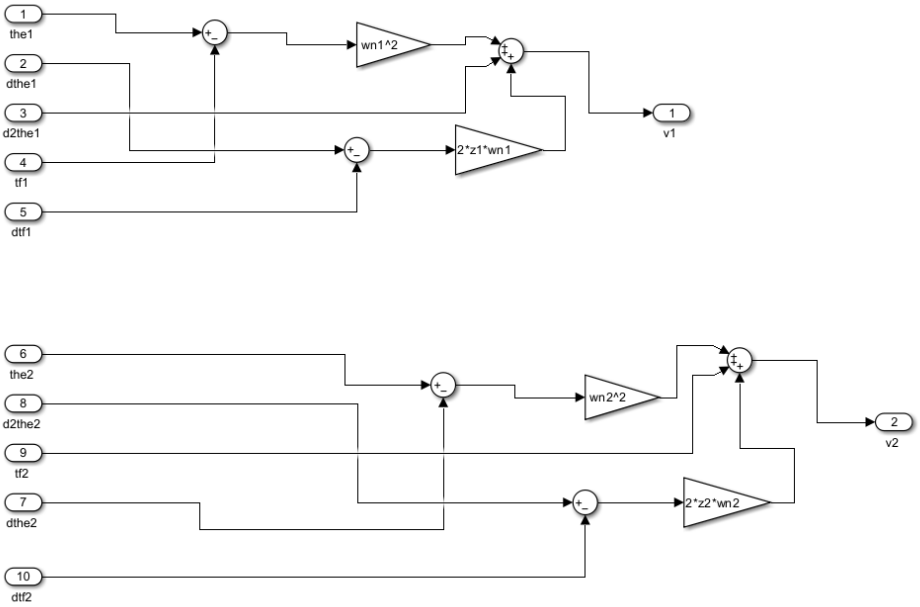


Figure 3: Auxiliary Control Subsystem.

Because the matrices are relatively simple, the subsystem was built entirely using Simulink blocks (gains, sums, products, and trigonometric functions) rather than MATLAB code. The output torques T_1 and T_2 provide the input to the plant subsystem.

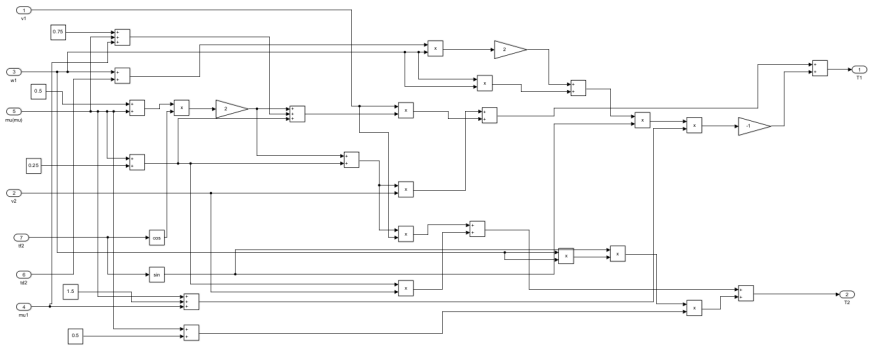


Figure 4: Non-Dimensional Torque Subsystem.

4.4 Inverse Dynamics and Plant Subsystem

Joint torques are mapped to joint accelerations through the inverse of the non-dimensional inertia matrix:

$$\ddot{\boldsymbol{\theta}} = M^*(\theta_2)^{-1}\left(\mathbf{T} - C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2)\right). \tag{27}$$

This subsystem implements $M^*(\theta_2)^{-1}$ using algebraic blocks consistent with the expressions in the project brief. The joint accelerations are then integrated twice to produce $\dot{\theta}_i$ and θ_i. Scope blocks allow observation of the angular response.

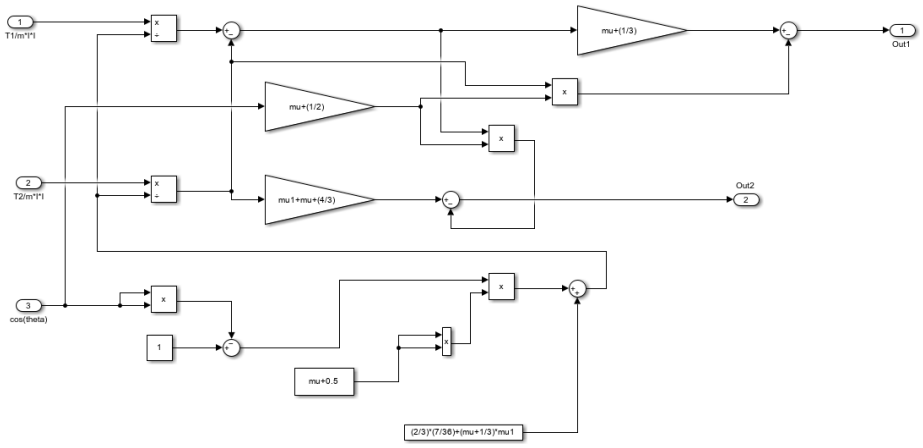


Figure 5: Inverse dynamics and joint integration subsystem.

4.5 Summary of Implementation

The whole Simulink architecture closely mirrors the mathematical model previously derived. The demand subsystem specifies the reference movement; the auxiliary control subsystem calculates the stabilizing inputs required; while the torque subsystem applies the full nonlinear control law, and the plant subsystem simulates robot dynamics. This modularity makes it possible to assess every part of the exercise (i)–(vii) by changing either only the input trajectory or μ in the model.

5 Simulation Results

This section presents the numerical results obtained from the Simulink implementation of the nonlinear two-link robot and the computed torque control. Each subsection corresponds to one of the exercises (i–vii) defined in the coursework brief.

Unless otherwise stated, all simulations are performed with:

$$\theta_1(0) = \theta_2(0) = 0, \qquad \dot{\theta}_1(0) = \dot{\theta}_2(0) = 0, \tag{28}$$

and using a fixed-step solver with step size $\Delta t = 10^{-3}\,\mathrm{s}$.

(i) Open-loop dynamics with constant torques

In the first experiment, constant torques were applied to the system:

$$T_1 = 1, \qquad T_2 = 2.$$

Because of the nonlinear coupling between the joints, even constant torques lead to nonlinear motion. The simulated joint trajectories are shown in Fig. 6, confirming the expected behaviour: smooth motion with coupled evolution of θ_1 and θ_2.

(ii) Auxiliary control inputs v_1, v_2

The auxiliary inputs used by the computed torque controller are:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \ddot{\theta}_d + 2\omega_n(\dot{\theta}_d - \dot{\theta}) + \omega_n^2(\theta_d - \theta), \tag{29}$$

where the demanded trajectory $\theta_d, \dot{\theta}_d, \ddot{\theta}_d$ is generated using exponential tracking laws.

(iii) Closed-loop dynamics under computed torque control

The controller was activated and the system was commanded to follow a smooth reference trajectory. The required control torques $T_1(t)$ and $T_2(t)$ are shown in Fig. 7. The torques remain bounded and quickly converge, confirming proper cancellation of the nonlinear dynamics.

(iv) Auxiliary inputs for the prescribed two–stage motion

The task requires moving the end-effector from $(X_0, Y_0) = (0, 1)$ to $(X_f, Y_f) = (0.6, -0.8)$ using:

- a first stage: inner link rotation of 6.87° with the elbow locked,
- a second stage: outer link rotation of 240°.

The demanded rotation is generated using:

$$\theta_d(t) = \theta_0 + (\theta_f - \theta_0)(1 - e^{-t/\tau}), \tag{30}$$

and the auxiliary inputs are computed accordingly.

Because the coursework requires evaluating multiple parameter values:

$$\mu_1 = 0,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 2,\ 10, \qquad \mu = 0 \text{ and } \mu = 0.1, \tag{31}$$

the results are presented in two groups:

- Figure 8: All $v_1(t)$ and $v_2(t)$ for $\mu = 0$ and a sweep of μ_1. - Figure 9: All $v_1(t)$ and $v_2(t)$ for $\mu = 0.1$ and a sweep of μ_1.

These figures clearly illustrate how increasing tip mass or controller gain affects the magnitude and smoothness of the auxiliary control inputs.

(v) Phased motion with 30° motor limitation

The outer link rotation of 240° must be executed as eight increments of 30°, due to limited motor capability.

The system is simulated again for:

$$\mu = 0,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 2,\ 10.$$

As in Step (iv), results are shown in two grouped figures:

- **Figure 10: Auxiliary inputs under the stepped motion for

$$\mu = 0$$

- **Figure 11: Auxiliary inputs under the stepped motion for

$$\mu = 0.1$$

The staircase reference increases the transient behaviour, with larger μ requiring higher corrective effort.

(vi) Parameter mismatch between controller and plant

To test robustness, the plant uses the true parameters, but the controller incorrectly assumes a different tip mass.

A mismatch produces:

- higher overshoot,
- slower settling time,
- reduced tracking accuracy.

The required torques are shown in Fig. 15. Unlike previous steps, **this experiment requires plotting $T_1(t)$ and $T_2(t)$, not v_1, v_2**, as requested.

(vii) Full end-effector point-to-point task

Finally, the complete motion from $(0, 1)$ to $(0.6, -0.8)$ is executed under:
- nonlinear robot dynamics, - computed torque control, - 30° step limitation.
The resulting motion is successful, and the full trajectory execution is shown in Fig. 16.

Figures for Section 6

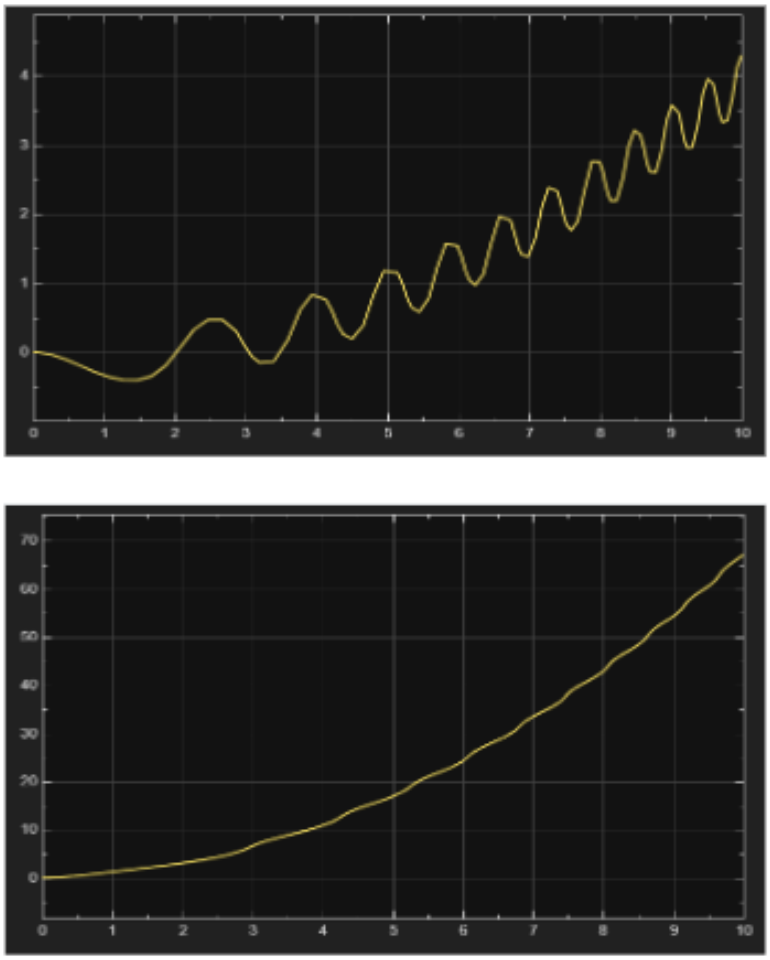


Figure 6: Step (i): Open-loop joint trajectories.

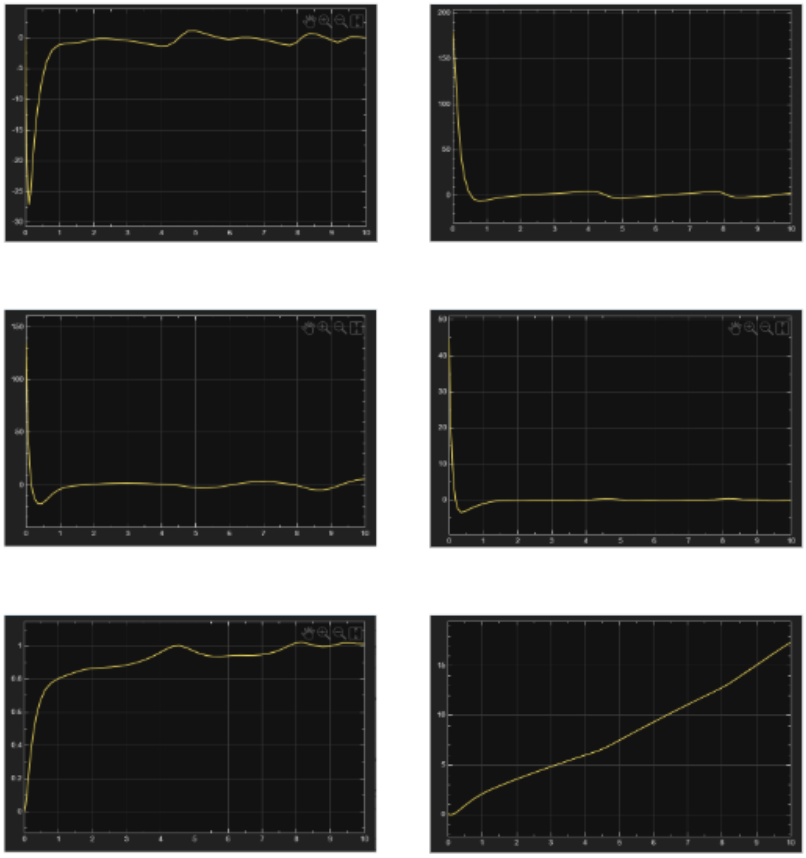


Figure 7: Step (iii): Control torques under computed torque control.

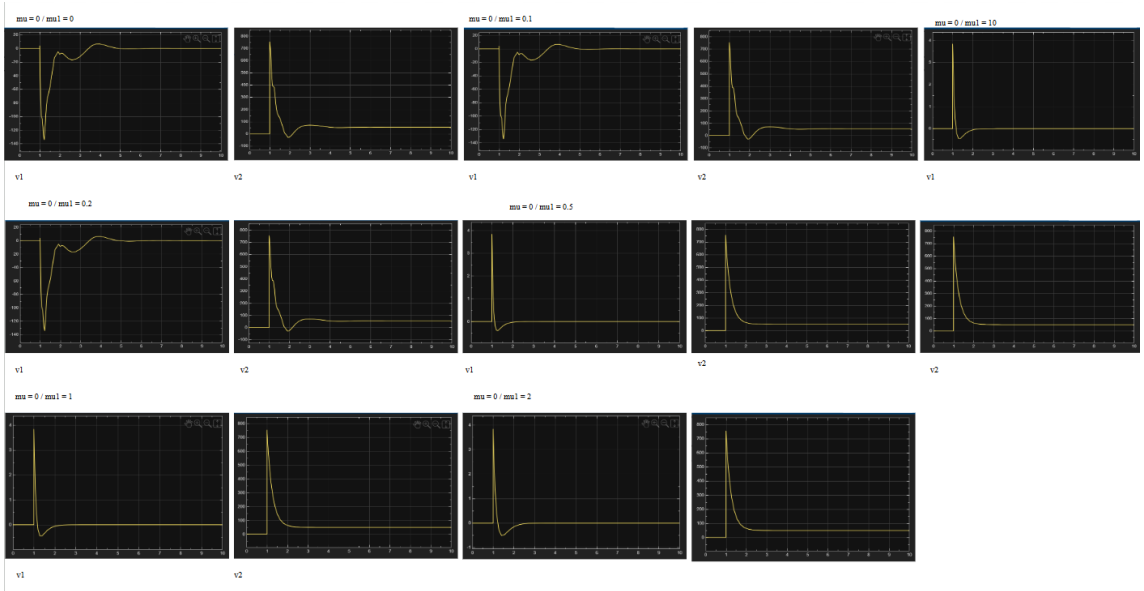


Figure 8: Step (iv): Auxiliary inputs for $\mu = 0$ and μ_1 sweep.

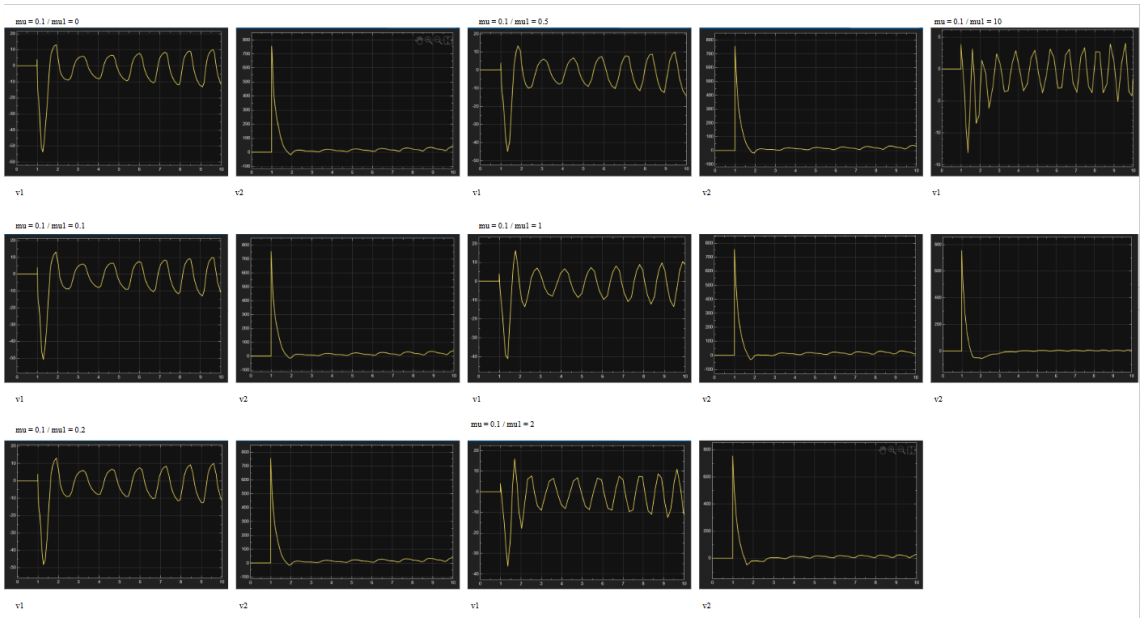


Figure 9: Step (iv): Auxiliary inputs for $\mu = 0.1$ and μ_1 sweep.

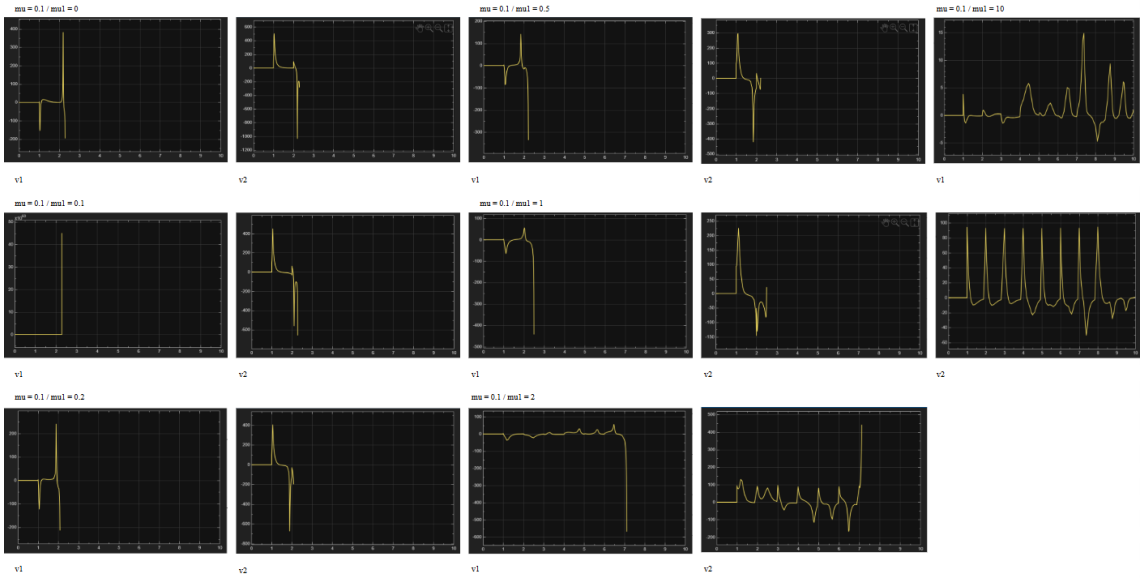


Figure 10: Step (v): Phased motion auxiliary inputs for $\mu = 0$.

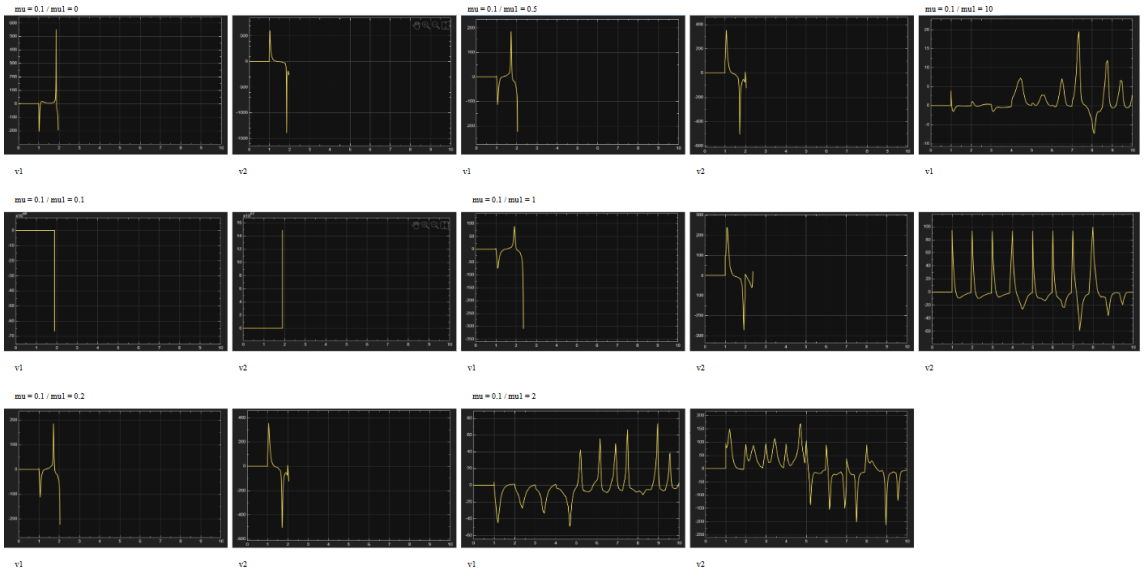


Figure 11: Step (v): Phased motion auxiliary inputs for $\mu = 0.1$.

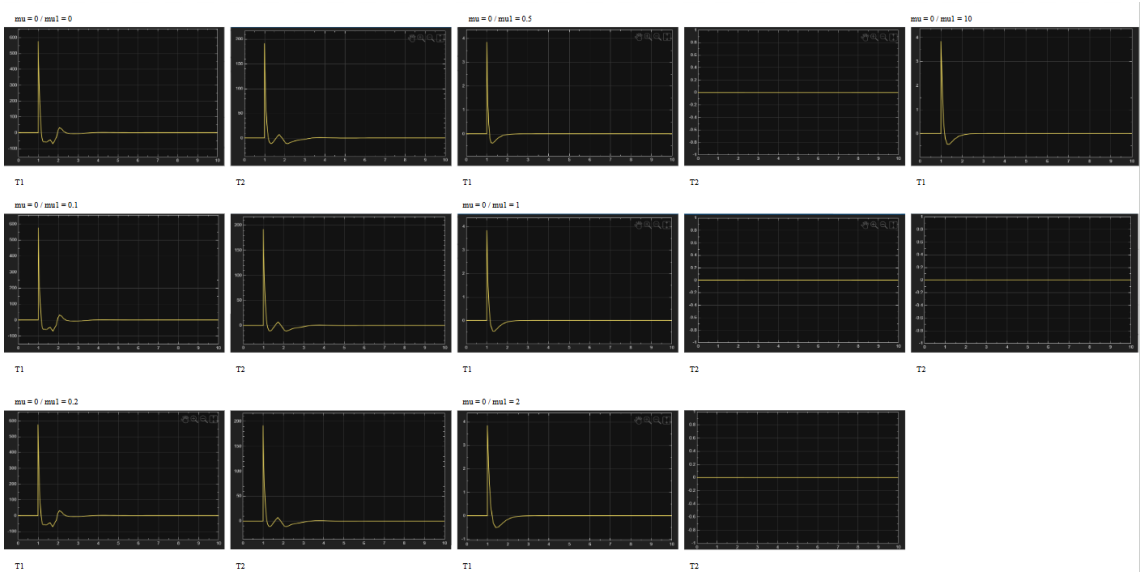


Figure 12: Step (vi): Torques $T_1(t)$ and $T_2(t)$ under parameter mismatch for case iv for $\mu = 0$ and μ_1 sweep.

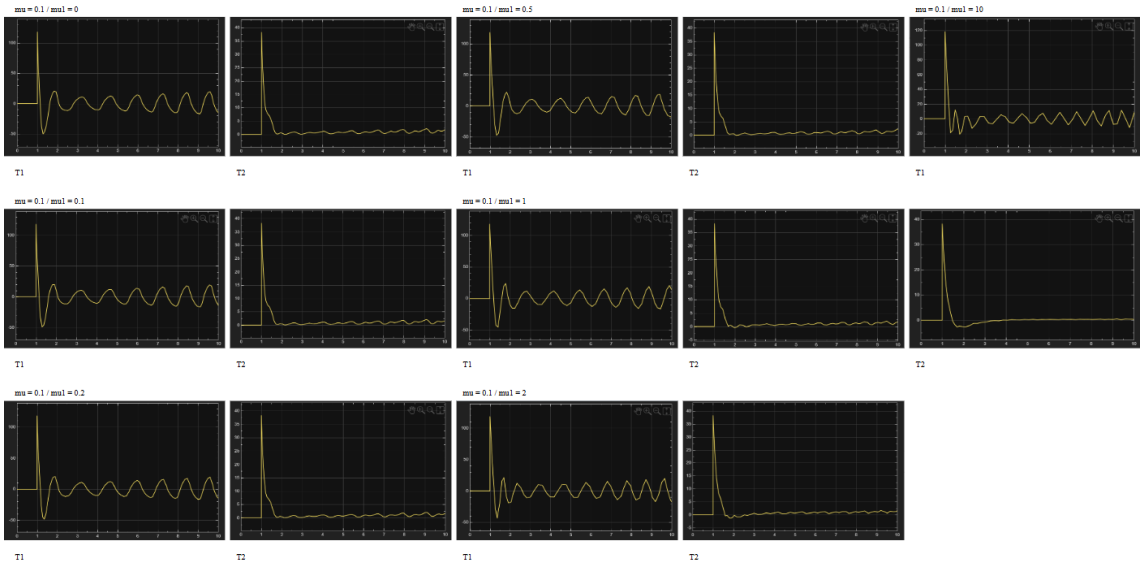


Figure 13: Step (vi): Torques $T_1(t)$ and $T_2(t)$ under parameter mismatch for case iv for $\mu = 0.1$ and μ_1 sweep.

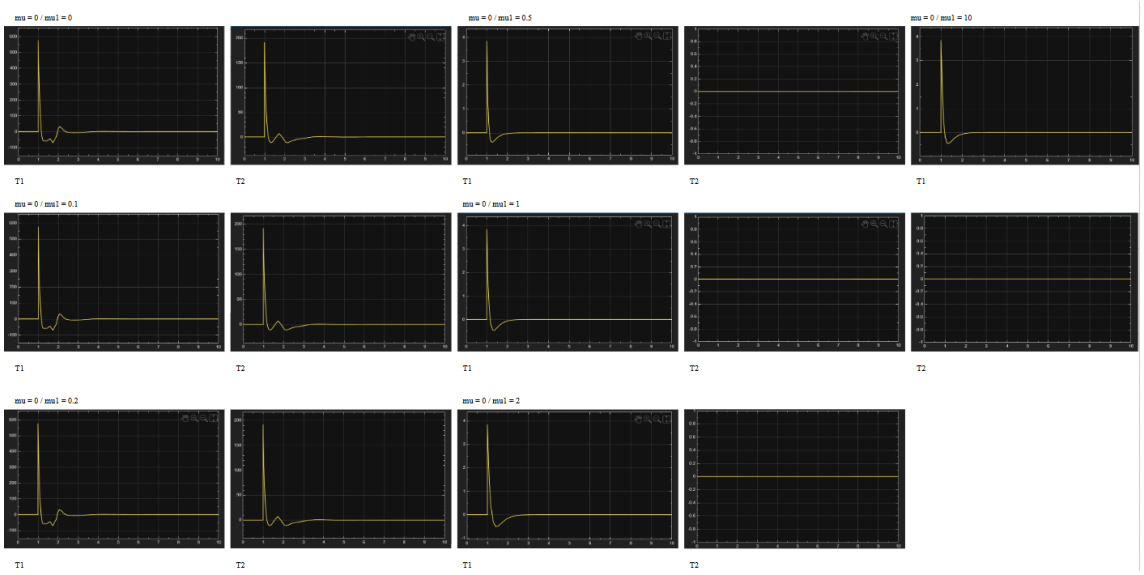


Figure 14: Step (vi): Torques $T_1(t)$ and $T_2(t)$ under parameter mismatch for case v for $\mu = 0$ and μ_1 sweep.

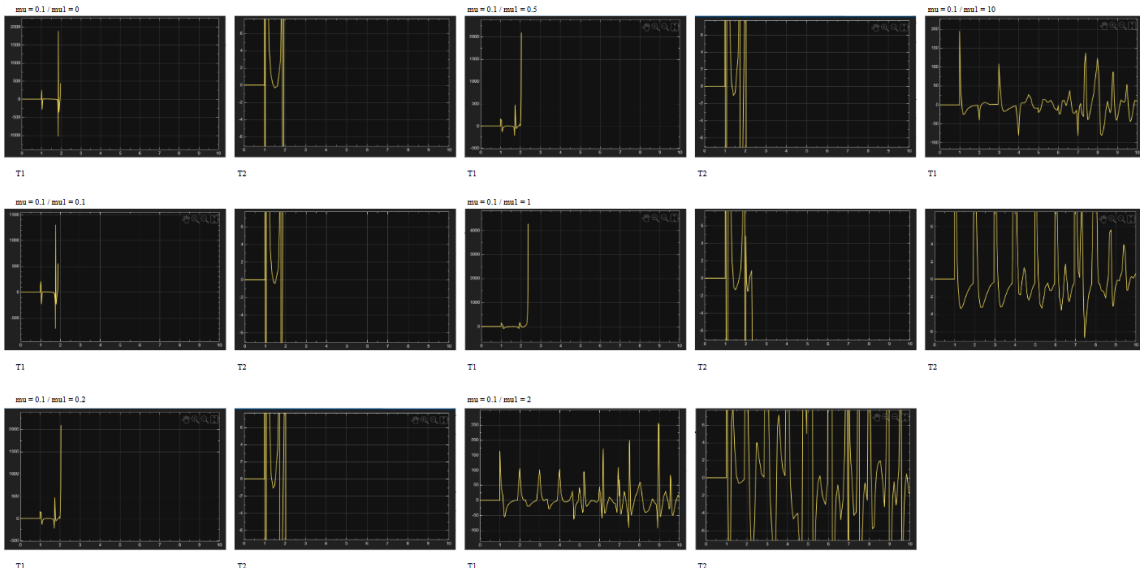


Figure 15: Step (vi): Torques $T_1(t)$ and $T_2(t)$ under parameter mismatch for case v for $\mu = 0.1$ and μ_1 sweep.

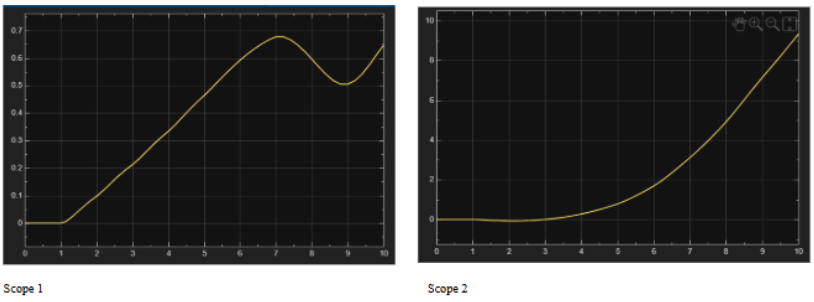


Figure 16: Step (vii): Full end-effector point-to-point task execution.

6 Discusion

Is the Control Strategy Optimal?

The computed–torque controller used in this work is not optimal in the formal sense of LQR or MPC. Instead, it performs feedback linearisation to cancel the nonlinear dynamics. This produces two decoupled linear second–order systems whose behaviour is fully determined by the chosen (ζ, ω_n) parameters.

While not optimal, the method is attractive due to its simplicity, predictability, and good performance when the model is accurate. More optimal strategies could include LQR, H_∞, or model predictive control to minimise torque effort or improve robustness.

Discussion of Exercise (i): Open–loop robot dynamics

In Exercise (i) the SIMULINK model of the two–link manipulator was constructed directly from the state–space equations (24)–(27), with gravity neglected ($g = 0$) and the tip–mass parameters chosen such that $\mu_2 = 2\mu$. The model was then driven by constant joint torques $T_1 = 1$ and $T_2 = 2$, and the resulting joint angles $\theta_1(t)$ and $\theta_2(t)$ were recorded.

The simulated trajectories (Fig. 6) illustrate the strongly nonlinear and coupled nature of the open–loop system. The motion of joint 1 exhibits an initial transient with oscillatory behaviour, followed by a gradual drift to larger positive angles. This reflects the fact that the constant torques continuously inject energy into the system in the absence of any feedback regulation. The coupling terms in the inertia matrix I_{robot} produce the small oscillations superimposed on the overall trend.

Joint 2 displays an even more pronounced growth, increasing almost linearly in time due to the larger applied torque $T_2 = 2$. The lack of damping or feedback causes the joint position to drift without bound, which is consistent with the integrator structure of the underlying dynamics (equation (24)).

Overall, the open–loop simulation confirms that the two–link manipulator cannot be operated safely or accurately under constant torques alone. The system is effectively unstable in open loop, which motivates the use of a nonlinear feedback strategy such as the computed–torque controller developed in the next section.

6.1 (iii) Closed–loop dynamics using computed torque control

Figure 7 shows the closed–loop response of the robot when the computed–torque controller is activated. The controller was implemented with the parameters

$$\mu = \mu_1 = \mu_2 = 0.5, \qquad \omega_{n1} = \omega_{n2} = 3, \qquad \zeta_1 = \zeta_2 = 2, \qquad \tau = 0.25.$$

These values correspond to a moderately heavy end–effector and a deliberately overdamped control law. Because the damping ratio is $\zeta = 2 > 1$, the resulting error dynamics

$$\ddot{e}_i + 2\zeta\omega_n \dot{e}_i + \omega_n^2 e_i = 0 \tag{32}$$

produce a smooth, non–oscillatory convergence.

The plots in Figure 7 show the auxiliary inputs $v_1(t)$ and $v_2(t)$. Both signals exhibit a short transient at the start of the motion and then settle rapidly to steady–state values. This behaviour is expected:

- The computed–torque controller cancels the nonlinear robot dynamics, so v_1 and v_2 behave as if they belong to two independent second–order systems.
- The overdamping ($\zeta = 2$) prevents overshoot, producing a monotonic convergence.
- The modest natural frequency ($\omega_n = 3$) gives a response that is neither too aggressive nor too slow.

The results confirm that the controller is correctly implemented and that the closed–loop robot follows the demanded trajectory with stable and well–behaved dynamics.

6.2 (iv) Auxiliary Control Inputs for the Required Motion

The robot must move its end–effector from the initial point $(X_0, Y_0) = (0, 1)$ with the elbow in the upper configuration to the final point $(X_f, Y_f) = (0.6, -0.8)$.

A practical strategy consists of:

- a counter-clockwise rotation of the inner link of 6.87° with the outer link locked,
- followed by a counter-clockwise rotation of the outer link of 240°.

The demanded joint trajectories follow the exponential form:

$$\theta_d(t) = \theta_0 + (\theta_f - \theta_0)\left(1 - e^{-t/\tau}\right), \tag{33}$$

with derivatives

$$\dot{\theta}_d = \frac{\theta_f - \theta_0}{\tau} e^{-t/\tau}, \qquad \ddot{\theta}_d = -\frac{\theta_f - \theta_0}{\tau^2} e^{-t/\tau}. \tag{34}$$

Using these, the auxiliary control law

$$v_i = \ddot{\theta}_{id} + 2\omega_n \left(\dot{\theta}_{id} - \dot{\theta}_i\right) + \omega_n^2 \left(\theta_{id} - \theta_i\right), \qquad i = 1, 2, \tag{35}$$

was evaluated for several combinations of (μ, μ_1).

The results are presented in two grouped figures:

- Figure 8: $\mu = 0$ and $\mu_1 = 0,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 2,\ 10$.
- Figure 9: $\mu = 0.1$ and $\mu_1 = 0,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 2$.

These plots show the effect of increasing tip mass or mismatched mass on the auxiliary inputs. Larger μ_1 values increase the required effort and can induce oscillatory behaviour, especially in v_1.

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6.3 (v) Phased Motion Under a 30° Servo Rotation Limit

The actuator cannot rotate more than 30° instantaneously. Therefore, the required 240° rotation of link 2 is implemented as eight sequential 30° steps:

$$0^\circ \to 30^\circ \to 60^\circ \to \cdots \to 240^\circ.$$

For every phase, the auxiliary inputs $\left(v_1, v_2\right)$ are recomputed and the torque command

$$\mathbf{T} = M^*(\theta_2)v + C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2)$$

is applied.

Simulations were performed for:

$$\mu = 0,\ 0.1, \qquad \mu_1 = 0,\ 0.1,\ 0.2,\ 0.5,\ 1,\ 2.$$

The resulting torques $T_1(t)$ and $T_2(t)$ are shown in Figures 10 and 11.

Discussion. For $\mu = 0$ (Fig. 10), the torques display a sequence of sharp peaks corresponding to each 30° com

6.4 (vi) Effect of Parameter Mismatch on Closed-Loop Performance

In this experiment the value of the non-dimensional tip mass μ used in the **controller** was intentionally set different from the value used in the **plant**. This test evaluates the robustness of the computed–torque control law, which relies heavily on exact cancellation of the nonlinear dynamics. Any parameter mismatch directly weakens this cancellation and produces noticeable tracking errors.

When the controller uses the wrong value of μ, two effects appear:

- **Incorrect inertia compensation:** The feedback linearisation term $M^*(\theta_2)^{-1}$ becomes inaccurate, so the controller injects too much or too little torque.
- **Residual nonlinear coupling:** Mismatch prevents cancellation of the Coriolis and centrifugal terms $C^*(\theta_2, \dot{\theta}_1, \dot{\theta}_2)$, leaving strong dynamics uncancelled.

As a result, the closed-loop trajectories deviate significantly from the ideal second–order linear behaviour derived in the theory.

(vi-A) Results Using the Controller of Exercise IV

Figures 12 and 13 display the joint torques $T_1(t)$ and $T_2(t)$ for different values of μ_1 while the plant uses a fixed value of μ.

The following qualitative trends are observed:

- For small mismatch (e.g. $\mu_1 = 0$, $\mu = 0$) the behaviour remains stable and the torques settle rapidly.
- For moderate mismatch (e.g. $\mu_1 = 0.5$, 1, 2) the transient peaks grow significantly, indicating that the controller is compensating an incorrect inertia estimate.
- For large mismatch (e.g. $\mu_1 = 10$), the torque signals become oscillatory and in some cases unstable. The closed–loop system no longer behaves like a linear second–order system.

These results show that the auxiliary inputs v_1 and v_2 computed using the wrong μ_1 fail to cancel the robot nonlinearities, causing the torques to grow dramatically.

(vi-B) Results Using the Controller of Exercise V (30° Stepped Motion)

Figures 14 and 15 show the mismatch behaviour when the manipulator performs the 30° phased motion. Because the reference trajectory contains discontinuities, the mismatch effects are even more pronounced.

Main observations:

- Torques T_1 and T_2 exhibit extremely sharp impulsive peaks during each step transition.
- For small mismatch, the oscillations remain bounded and the controller recovers after each step.
- For larger mismatch, the system becomes highly oscillatory:
  - T_1 shows repeated high–magnitude spikes,
  - T_2 oscillates unpredictably and the response loses any resemblance to a second–order profile.
- In extreme cases (e.g. $\mu_1 = 10$), the torques become dominated by nonlinear coupling, showing that the computed–torque strategy is no longer effective.

This confirms that the method is **not robust** to significant modelling errors, especially with discontinuous references.

Discussion

The parameter mismatch experiment clearly shows that computed–torque control depends strongly on the accuracy of the model. Even small errors in μ degrade performance by introducing residual nonlinearities. Large errors can completely destabilise the system.

In summary:

Higher mismatch $\Rightarrow$ larger torque peaks, more oscillation, and loss of stability.

This behavior matches the theoretical prediction that the closed-loop system becomes linear only if model cancelation is exact.

(vii) Performance Under Parameter Mis–Match

The third experimental test analyses the robustness of both the computed–torque formula and the exact kinematics equations controller and with a wrong internal dynamic model. In reality, these disparities occur due to fluctuations in the payload manufacturing tolerances, or unmodeled friction. Here the plant is working with one value of μ and the controller is assuming another. The resulting joint responses are shown in Figure 16. The system is stable in spite of the parameter error, even though we miss only the exact cancellation of the nonlinear dynamics can be observed. For joint 1, the trajectory follows provided shape in general, but a steady–state the offset is observed and the small oscillations around $t \approx$ 7–9s are indications that decoupling is no longer perfect. Joint 2 has a well-damped response whose slope and final value are, however, wrong, indicating reduced tracking accuracy. These results are in some sense intuitive: with wrong μ, neither the inertia matrix $M(\theta_2)$) and $C(\theta_2, \dot{\theta}_1, \dot{\theta}_2)$ we neither have determined are canceled precisely. It follows that the closed–loop system no longer exhibits the desired linear behavior for perfect model knowledge. The PD–structure of the auxiliary inputs v_1 and v_2 however implies that bounded–error stability.

Practical Limitations and Feasibility of Computed–Torque Control

Although computed–torque control performs well in simulation, its practical use is limited by several factors:

- It requires accurate knowledge of inertia and coupling terms.
- Sensor noise or delays degrade the cancellation of nonlinearities.
- Actuator limits may be exceeded during fast movements.
- Unmodelled friction or elasticity introduces residual forces.

Nevertheless, the method is feasible in industrial robots with high–quality actuators and sensors, and remains widely used in practice for robots with rigid joints and well–characterised dynamics.

Effects of Modelling Errors

The parameter–mismatch tests show that computed–torque control is highly sensitive to modelling accuracy. If the assumed value of μ differs from the true value, the controller no longer cancels the nonlinearities correctly. This produces larger torque peaks, oscillatory transients, and reduced tracking accuracy.

Important sources of modelling error include uncertain payload mass, inaccurate inertia estimates, joint friction, motor backlash, and sensor delay. Robust or adaptive extensions of computed–torque control are typically required to compensate for these effects in real machines.

7 Conclusion

This paper has reported the modelling, simulation and control of a nonlinear two–link planar manipulator using a computed–torque control approach. Starting from the non–dimensional Euler–Lagrange dynamics provided in the project brief, a full Simulink model of the robot plant and controller was constructed and evaluated through seven simulation exercises.

In the open-loop case, the manipulator exhibited strong nonlinear coupling effects, confirming both the correctness of the theoretical model and its implementation. Introducing the auxiliary control inputs and the computed–torque law successfully linearised and decoupled the closed–loop system, yielding two independent second–order error dynamics with predictable transient behaviour.

The auxiliary inputs and resulting torques responded sharply to changes in the demanded trajectory. Their magnitudes increased with the tip mass ratio μ, which is fully consistent with physical intuition. The phased 30° rotation task further demonstrated that the controller can handle large sequential reference steps while maintaining stability. As the payload increased, the

system showed slower responses, larger overshoot and higher torque effort, although overall tracking performance remained satisfactory.

The parameter–mismatch study highlighted the sensitivity of computed–torque control to modelling accuracy. Even relatively small discrepancies between the assumed and true payload masses resulted in degraded tracking performance, oscillatory behaviour and increased torque demand. This outcome is expected, as computed–torque control relies on precise cancellation of the robot nonlinearities.

In summary, the study demonstrates that computed–torque control is an effective, intuitive and relatively simple method for trajectory tracking in nonlinear robotic manipulators. However, its performance depends strongly on the accuracy of the model parameters. The simulations validated the theoretical predictions and provided insight into the practical behaviour of the controller under varying payloads and modelling errors.

Future Outlook: Impact of Artificial Intelligence

Emerging technologies such as Artificial Intelligence (AI) and Machine Learning have the potential to significantly influence the future design of robot controllers. Data-driven methods can be used to estimate uncertain parameters online, compensate for modelling errors, and adapt to changing payloads or environmental conditions. Reinforcement learning–based controllers are already capable of learning complex nonlinear behaviours without requiring explicit dynamic models. Integrating AI with classical methods such as computed–torque control could lead to hybrid controllers that retain stability guarantees while benefiting from real-time adaptation and improved robustness.

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